The Robot Umpire Effect

Human Accuracy vs Machine

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Who will fans get mad at now?

MLB is currently testing Automated Ball-Strike (ABS) systems.

The "Human Element" is controversial: Tradition vs. Accuracy.

Goal: I wanted to build a "Ground Truth" model to measure exactly how many calls humans miss and *why* they miss them.

Research Questions

- **The Accuracy Gap:** Can a machine learning model outperform human umpires using only pitch location data?
- **Systematic Error:** Do humans miss calls randomly, or are there specific "blind spots"?
- **Height Bias:** Does the strike zone bias change based on the physical size of the batter?

Building the Pipeline

Source: MLB Statcast Data (via pybaseball).

Range: Summer 2025 (June 1 – Aug 31).

Volume: ~178,000 called pitches.

Cleaning: Performed 100% in Python (No Excel). I filtered swinging strikes, handled nulls, and normalized umpire names.

The "Ground Truth": I engineered a geometric label based on the official rulebook:

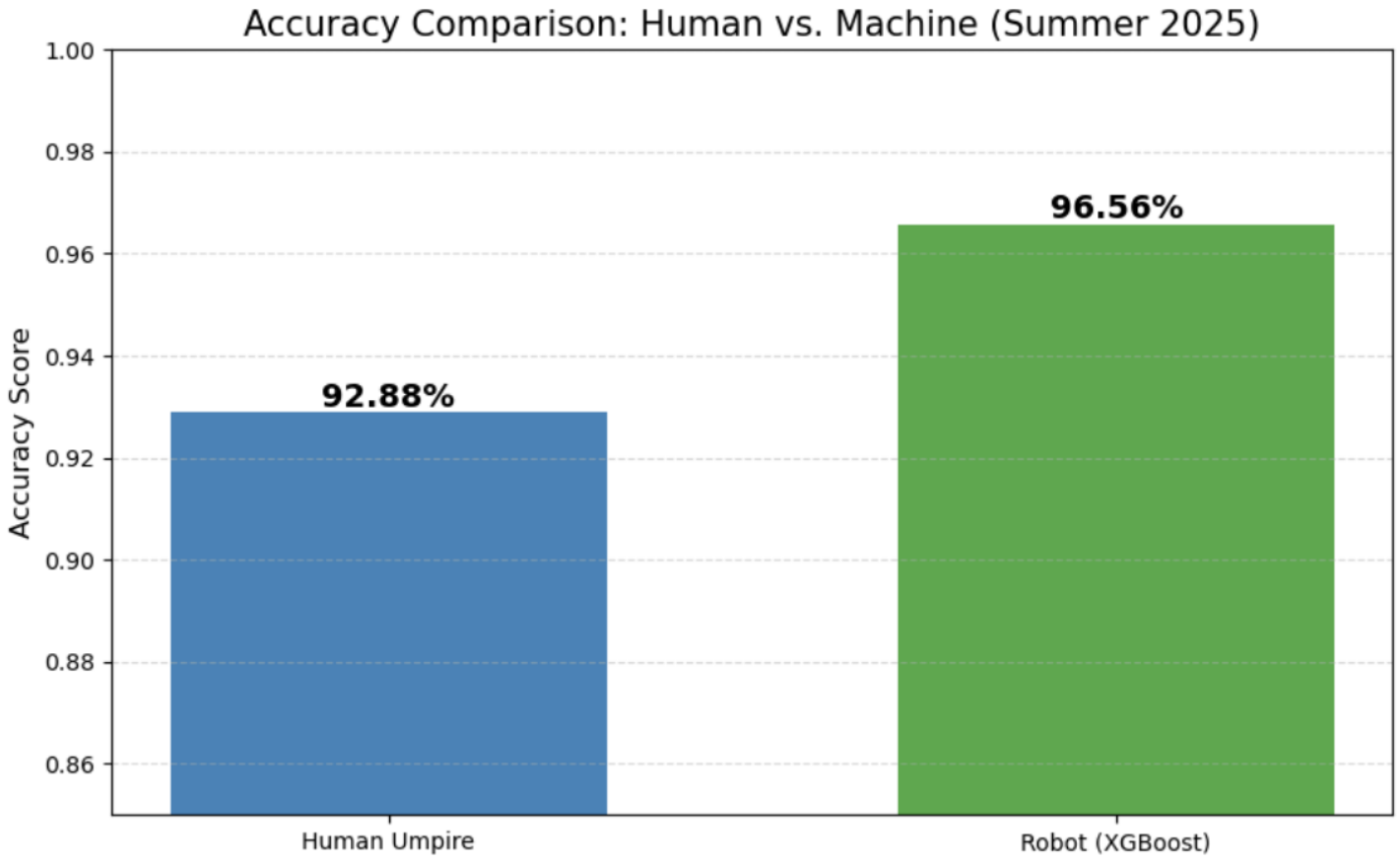
Width = 17" Plate + Ball Radius.

Height = Dynamic (Knee to Chest per batter).

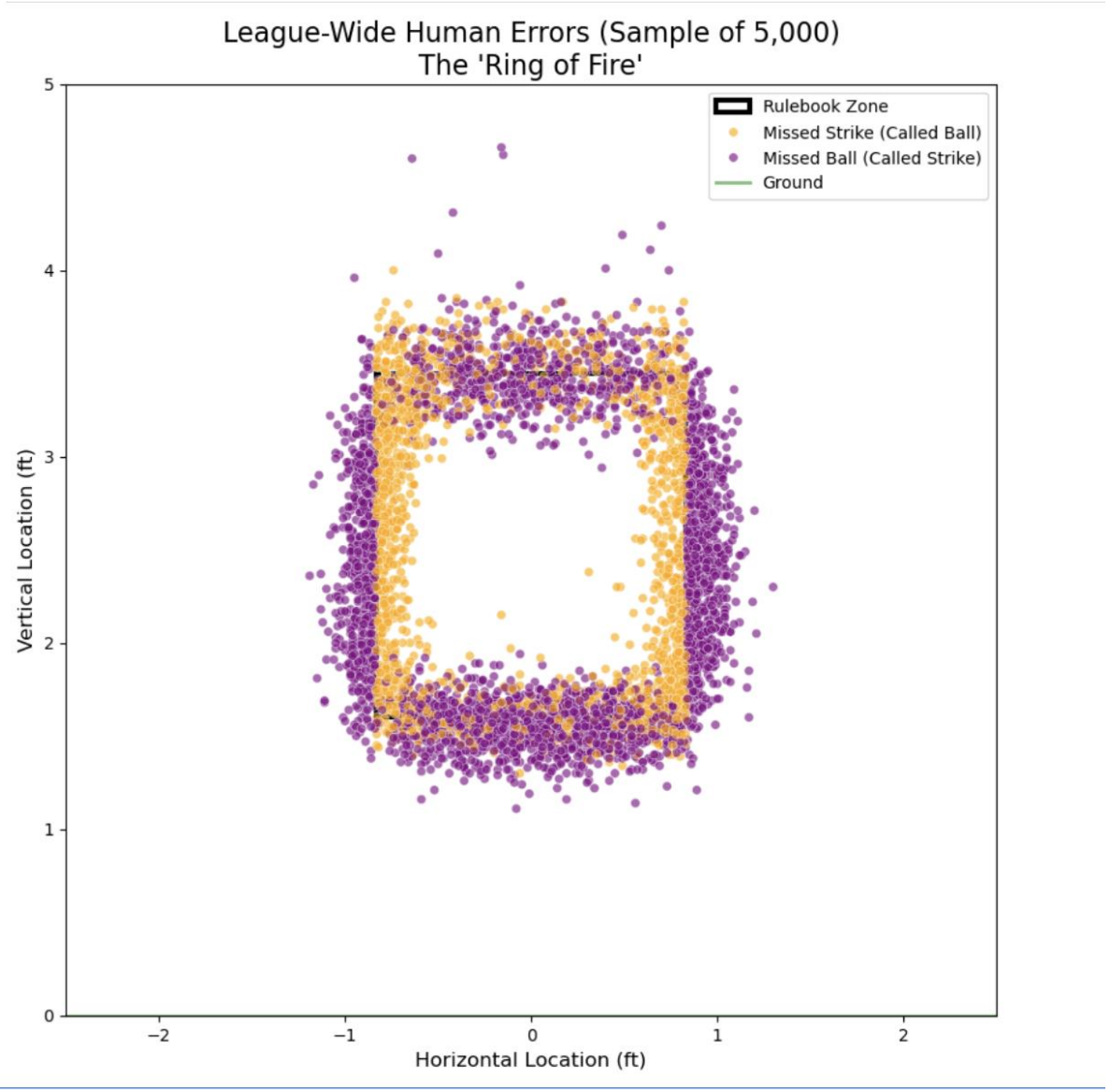
Modeling

- XGBoost Classifier
  - **Type:** Gradient Boosted Decision Trees.
  - **Inputs (Features):** plate_x, plate_z, release_speed, handedness.
  - **Target:** The Geometric "Ground Truth" (Rulebook Strike).
  - **Training:** 70/30 Train-Test split.

Accuracy Comparison. Who wins?

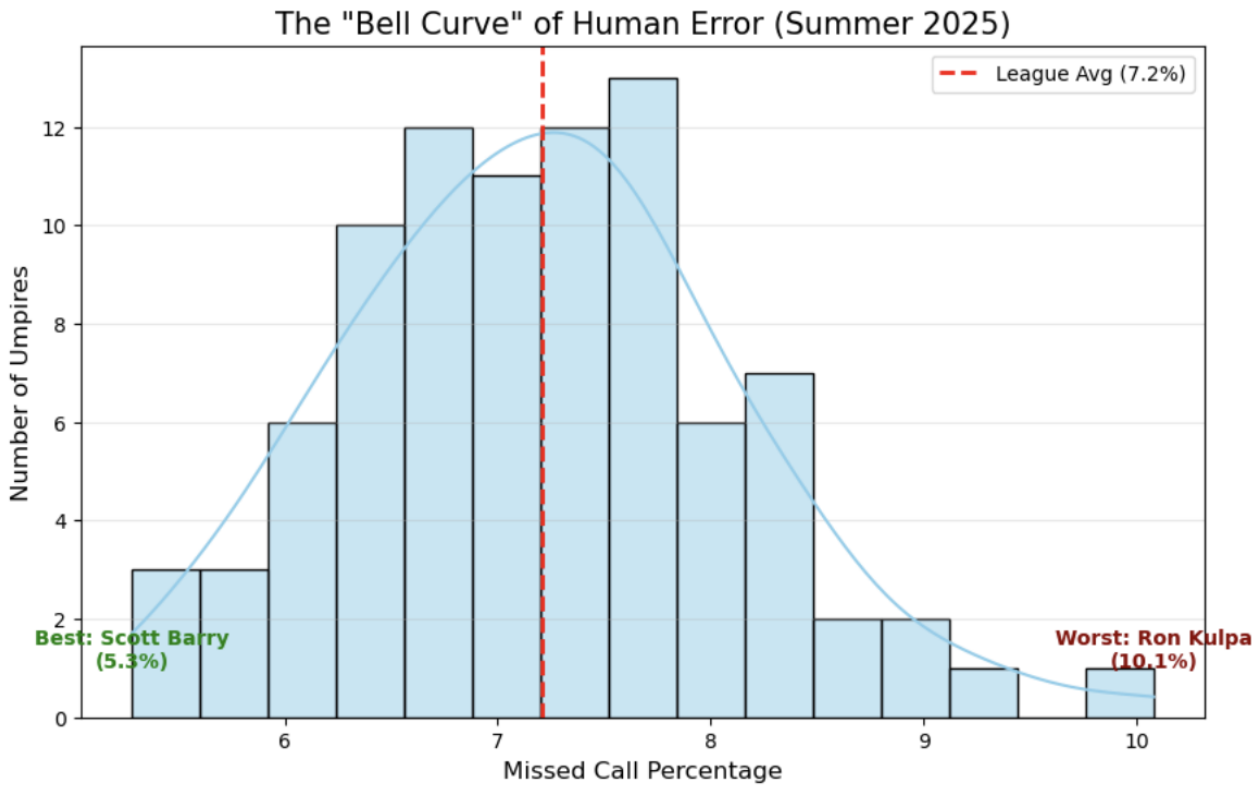


The “Ring of Fire”: Human Error



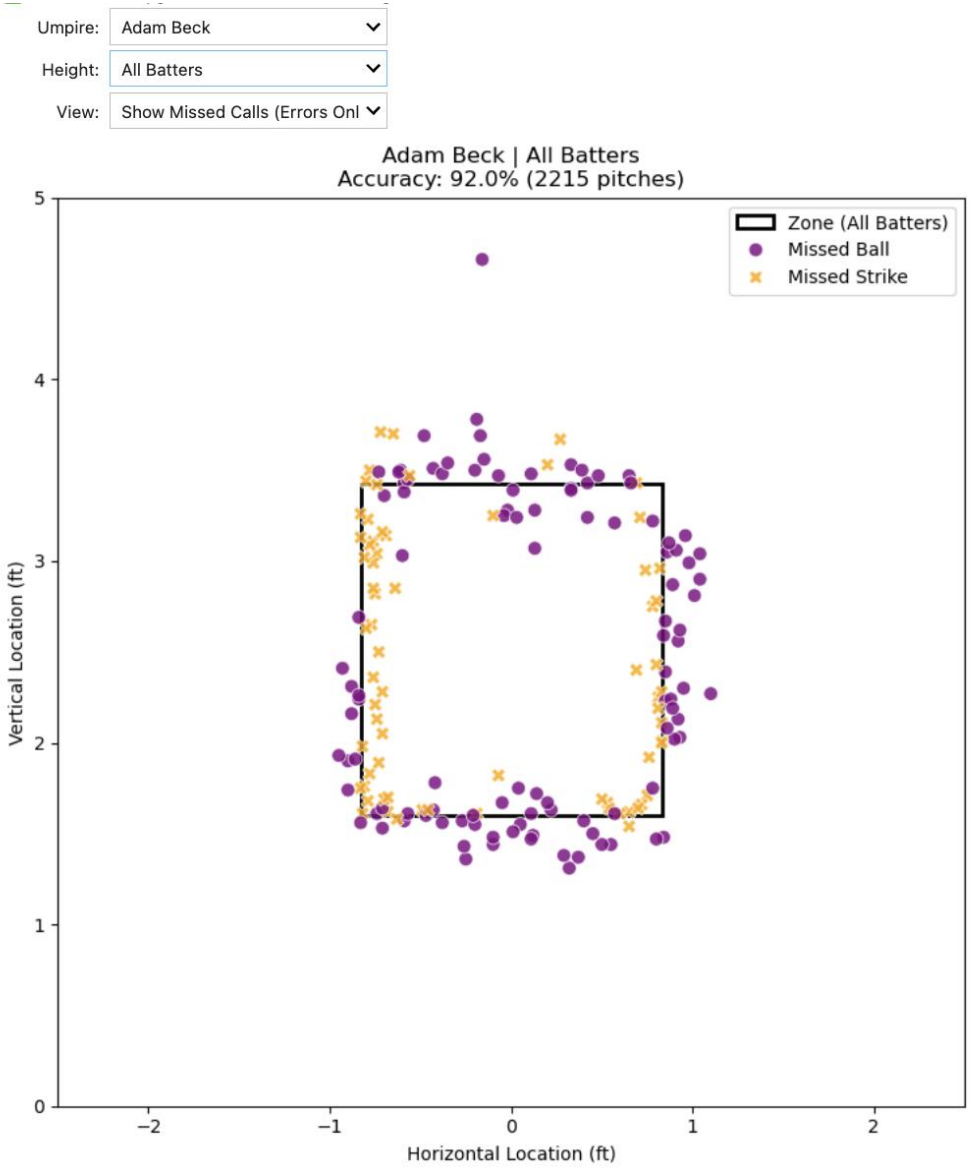
The "Bell Curve" of Human Error

- League Average Error: **~7.2%**
- **The Gap:** The worst umpires are nearly **2x** as error-prone as the best.
- **Key Finding:** Player experience depends entirely on *who* is officiating.



Real-Time Error Analysis

- Built using ipywidgets to enable real-time filtering of 178,000+ rows of data
- Instantly segments data by Umpire, Batter Height, and Call Accuracy without reloading the dataset.
- Visually proves that the model's "Ground Truth" logic correctly adjusts to the physical reality of the batter.



Limitations & Technical Nuance

- **Model vs. Reality:**
  - **Real ABS:** Deterministic. Hard-coded geometry (Rulebook is absolute).
  - **My Model:** Probabilistic. Learned patterns from training data (Approximation).
- **Why not 100% Accuracy?**
  - The model achieves **96.56%**, not 100%.
  - Gradient Boosted Trees must *infer* the box boundaries, whereas strict code *defines* them.
- **Data Constraints:**
  - Relies on raw Statcast tracking (which has its own margin of error).

The Verdict: Robots vs. Humans

Accuracy: Robots are statistically superior (+3.7% improvement).

Fairness: Humans introduce systematic bias (The "Ring of Fire") and inconsistency (The "Bell Curve").

The Consequence: Switching to robots would strictly enforce the rulebook, potentially shrinking the effective strike zone and changing how the game is played.